April 30, 1973

Mr. A. G. Moore

Weekly Report - April 16-27, 1973

Project 2903 - Service to Manufacturing

Use of Concentrated G-7 Extract

Some spotting by extract was noted on all G-7 sheet made while using extract from the aeration evaporator containing 24 to 32 percent solids* for the wet end extract sprays. However, a smoking evaluation on 100 percent G-7 cigarettes indicated that the spotting which occurred when using a 25 percent extract was not detrimental.

Three running periods of 13, 4 1/2, and 4 1/2 hours, respectively, using concentrated extract on the wet end of No. 4 Dryer have been made during an interval of about eight days. The first two test runs were made at dryer speeds of 300 to 310 ft./min. using 7 0.101-in. dia. sprays at 21-22 psi. although during the whole test period the wet pulp sheet was not very satisfactory and @ 3.8 gm./sq. ft. sheet had to be used. The third test run was made using 10 0.0935-in. dia. sprays at 18 psi. and @ 315 ft./min. dryer speed which was the maximum obtainable with present No. 4 dryer and wet machine speed controls. These speeds compare with @ 285 ft./min. speed normally used during the same period and indicated production rate increases up to 11 percent obtainable by using the extract of 25% solid content at the dryer inlet.

With this increase above the normal 20% extract it was possible to increase the proportion of solids applied at the dryer inlet from 20% to 35% of the total extract solids applied to the sheet. Product had satisfactory flavor and moisture content but was slightly heavy because of use of the heavier pulp sheet. Sheet shrinkage was about the same as for normal operation and no problem was experienced in using a can of the test run tobacco set up to contain 14.3 percent moisture and stored for 2 days.

Additional tests are planned using a more concentrated extract on the feed end of the dryer.

*(Note) All extract solid concentrations reported here were determined on the basis of Development Lab moisture analyses which show a 3 to 5 percent lower solids content than do the plant control tests. The possibility of changing the plant test method is being considered.

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